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Engineering 95

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Sensing the World Reflection #1

Done this week:

In this section of the module, we focused on the materials science and engineering aspects behind computer engineering. We were tasked with fabricating our own interdigitated electrodes (IDEs). A large focus was placed on learning the techniques used in research and industry to create chips. After fabricating our IDEs, we deposited conductive material such as silver paste. We measured the length, thickness, and resistance of the conductive material so the resistivity could be calculated. We investigated different effects on conductivity, such as moisture, which was done by breathing onto the IDE while measuring with the multimeter.

Before completing this work ourselves, it was important to gain knowledge of how this process is performed in research and in industry. We learned about the process of chip fabrication during our tour of the cleanroom facility. There, we were shown a demonstration of spin-coating, a technique that evenly distributes a thin layer of photoresist onto a substrate, which in this case was a silicon wafer. We learned that the photoresist is used to transfer a geometric pattern onto the substrate using lithography. Conductive metals are then deposited onto the substrate using metal-organic chemical vapor deposition (MOCVD). Equipped with this knowledge, we began our own fabrication process. A trained staff member at Wilbur used the vinyl cutter to cut out the Kapton film. A trained student then used the laser cutters to create the pattern on the Kapton film. With our freshly made IDEs, we went to Whitaker to begin the process of deposition. We

split into groups to perform the rest of the project. We first transferred our Kapton films onto glass slides using the backing tape. Some IDEs were made by cutting a measured rectangular area out of Kapton tape adhered to a glass slide. To deposit the metallic material, we applied either silver paste, silver paint, or carbon paint on top of the IDE, and then used a secondary glass slide to scrape off the excess and leave the remaining material deposited in the cavities. This technique is known as screen printing. We then dried the material onto the glass slide using a hot plate. Finally, we measured the resistance using the multimeter. We measured the length and thickness of the material. This allowed us to calculate resistivity using the equation $R = \frac{\rho L}{A}$. Solving for resistivity gives $\rho = \frac{RA}{L}$. Additionally, we applied PEDOT-PSS to the IDEs using spin-coating and repeated the previous steps to find resistivity.

From this module, I learned a lot about the process of chip manufacturing, IDEs, materials science, and the research labs at Lehigh. It was exciting to learn the processes used in industry and then get to apply them in real life. I learned the importance of material choice in engineering, where the properties of the material are important for the specific application.